Information-based Interactive Services and Support System

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***Abstract*—The fact that information-based interaction designed requires user involvement, Service Desk System (SDS) is necessary for collecting, tracking and processing requests for IT service support. The system can be dynamic, where many approaches can be used depending on the objective of the tasks at hand. Currently, Kulliyyah of Information and Communication Technology (KICT), International Islamic University Malaysia (IIUM) SDS do not cover a lot of evolving user's tasks. As a result, the main aim of this study is to propose interactive SDS for students and staffs at KICT, IIUM. The architectural framework of the system has been formulated to include all the necessary requirements that might have arisen over time. There are some drawbacks in the process of the current system which affects the whole management process of the organization. With the aid of the framework, a prototype has been implemented. This prototype is established using the Hypertext Markup Language (HTML), PHP with the support of Cascading Style Sheets (CSS3), Bootstrap and JavaScript along with MySQL as the database. . The developed system is an online (web-based) system which allows users to request information in order for the administrator to respond promptly, which shows the efficiency and reliability of the system.**

***Keywords*—users request; service desk; administrative response; online tasks**

I. INTRODUCTION

Service Desk System (SDS) is a management task assisted system, which provide support for effective workflow control through collecting, tracking and processing requests for IT service support processes within a company [1]. The system serves as a platform for connections between end users of IT services and the IT support staffs. This makes it part of the extensive process of establishing common IT standards within any business applications for ensuring strong quality and greatest efficiency of the IT operations [2]. Due to the nature of the complexity of Information technology services, the service desk is required to make a platform of communication between the internal IT provider and its users [3]. This platform can make incidents and problems to be quickly examined and solutions to be provided. One of the major roles of the system is to provide, an unforeseen report on operating scenarios regarding decisions to bring the improvements to meet the business demands.

As tasks are readily available on the platform of advanced technology, computerized and automated systems are comprehensive to serve the jobs within the organization, education and business factors. This has become possible because technology has a great ability and chance to reduce the burdens that are faced by the current system as well making it easier and manageable. In addition, the use of computerized systems in everyday life could increase the work performance to be more competent, accurate, consistent and reliable to suit with the current era that we live in.. However, many companies still prefer to choose a manual system to do their work. For instant, KICT service desk system that is currently manual.

For this reason, there is a need for providing a source of positive perks for a new system.. In pursuit of this, the current study reviewed the existing SDS and research studies related to the system [4-8]. Many SDS applications are available online. They provide different sort of services for an organization. Yet, it's necessary for SDS to cover the demand of any organization. Therefore, this study deemed it fit to propose SDS that would be able to provide the services offered by KICT. In the proposed system, the result of the first preliminary study shows that for every application form received by the KICT administrator, the record is done manually, from the form into the system. This pose a lot of problem where a staff might key in student's details wrongly, information might be incorrect and sometimes forms might be lost. Due to this matter, several problems may arise: The application forms are submitted in hard copy, facing the risks of unrecorded data or data redundancy, inadequate communication medium and slow response time. This creates time wasting for some unnecessary actions such as typing the same data over and over againand it would be difficult to be resolved within a specific time.

The objective of the proposed system is to convert the current task which is a paper based into a computerized system So as to reduce time-consuming that the user faced currently. The system is user-friendly effective. As a result, this study formulated the following specific objectives: To design a system that process student's application within a specific short period of time. To design a system that give privilege to students to do their application throughout the campus. To design a system that give privilege to staffs to complete all the process of students' application smoothly. To propose a technique that improves the quality of administrative service.

The remaining part of the paper contains 3 sections. Section 2 provide the system architecture and designed. Section 3 provide the evaluation of the system. Section 4 is conclusion.

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II. SYSTEM ARCHITECHTURE

This system is built on a concept of an online web-based application system. Online web-based application means that the codes, database and design of the whole system reside on a server and available all over the world as long as the users are on the internet. The whole data transaction process only occurs on the server itself. The reason for this design infrastructure is based on the concept of availability that the internet is offering and the fact that everyone can easily access the system with no hassle such as students and staff. The proposed development stages are presented in the flowchart in Figure 1:

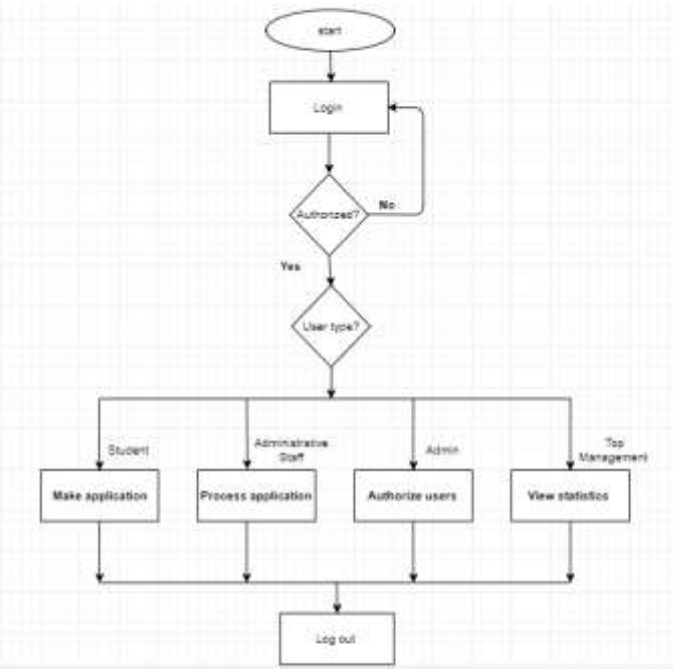

Fig 1. The flowchart of the proposed system

A. System Development Process

Creating new software or software-intensive systems is still a challenge [9]. The reasons lie with the increasing data required in a typical systems and the emergence of very large highly distributed systems. This study design of the proposed system uses the generic software development process. Activity diagram, use-case diagram, package diagram, class diagram, sequence diagram, and user interface design are the main requirements guiding the design of the proposed system. The analysis phase of this system answer the questions of who will use the system, what the system will do, where and when it will be used, during analysis phase data are collected on the available files, decision points and transactions handled by the present system.

The design phase of this system dwells on how the system will operate, in terms of the hardware, software, and network infrastructure; the user interface, forms, and reports that will be used; and the specific programs, databases, and files that will be needed. Using the requirements, it makes it easier for the user to understand the flow of the system, how the actors will communicate with objects, how activities will be conducted and how the system will function. Most importantly it assists for later review upon the completeness, correctness, and the consistency. A system development methodology means the framework is developing an information system that plans, structure and controls the process step by step [10]. One of the popular software development processes that combine both design and stages of prototype elements are the Spiral Model. It is built to integrate the benefits of top-down and bottom-up concepts. Spiral Development Model can also be called Spiral Lifecycle Model and this is because Systems Development Method (SDM) is intended tacke huge and costly projects. Ideally, the Spiral method has four major stages, namely: Planning, Risk analysis, Engineering, Customer evaluation.

While the requirements analysis is gathered during the planning phase. It's a critical task to fit requirements into 'BRS' that is 'Business Requirement Specifications' and 'SRS' that is 'System Requirement specifications'. This is based on the intersections of some views and facts. In the risk analysis phase, a process is undertaken to identify risk and alternate solutions. If any risk is found during the risk analysis, then alternate solutions are suggested and implemented.

After evaluation phase, it is considered that the software is developed, and the testing is done. In this phase, the software is offered to the user to evaluate and give feedback on the project before the project continues to the next phase. There are many advantages of Spiral model: highly focused to avoid risk. This model is good for critical and larger projects where avoidance of risk is highly prioritized. Another advantage is to add functionality at a later date. In this software life cycle, software is developed at the earlier stage.

A requirement specification narrates the flow of the system and it also covers a set of use cases that explain the user interaction with the software [11]. There are two kinds of requirement specifications for a software system: functional requirement define the functionalities and it is detailed in system design; non-functional requirement explain how a system should be and it is detailed in system architecture [12].

The Functional Requirements are; based upon the user requirements that the system should be displayed in a relevant interface without any errors. All the members of the site must be able to access the site without any problems at any time. The admin should be able to add any data as long as the admin falls under a set of rules and requirements, without any problems. The admin should have full access to the additional functions of the system without any hindrance. The system should make sure the access to the different functionalities is well controlled and managed, it should make sure that all the functions and privileges in the system are maintained at all times [11-12].

The Non-Functional Requirements are: The clients must be able to access their accounts 24 hours a day, seven days a week. The system should be able to record and receive all kinds of information from any forms and should update all aspects of the database without any errors. The customers should be notified through system interface announcement about any sort of maintenance of system, so as to avoid inconveniences to clients. The system interface should be dynamically updated by the admin in order to provide convenient information about the system services [12].

B. Logical Design

The design phase of the system ensure that the system is designed with all functionality that the user requires. In fact, the design stage is the structure of the system menu, for identifying any or all system objects which determines operations and data structures for all objects, whereby validating relationships and interactions between objects and user interface object prototypes.

1) Activity Diagram

Activity diagram prepares documentation for processing and also explain the process briefly through symbols and text.
It makes it clearer to define and perceive the steps in a process. Furthermore, it makes it easier to visualize vividly and helps to analyze and effect changes to improve the process [13] [12]. An activity diagram shows the overall flow of control. The activity diagram for this study is represented in Figure 2.

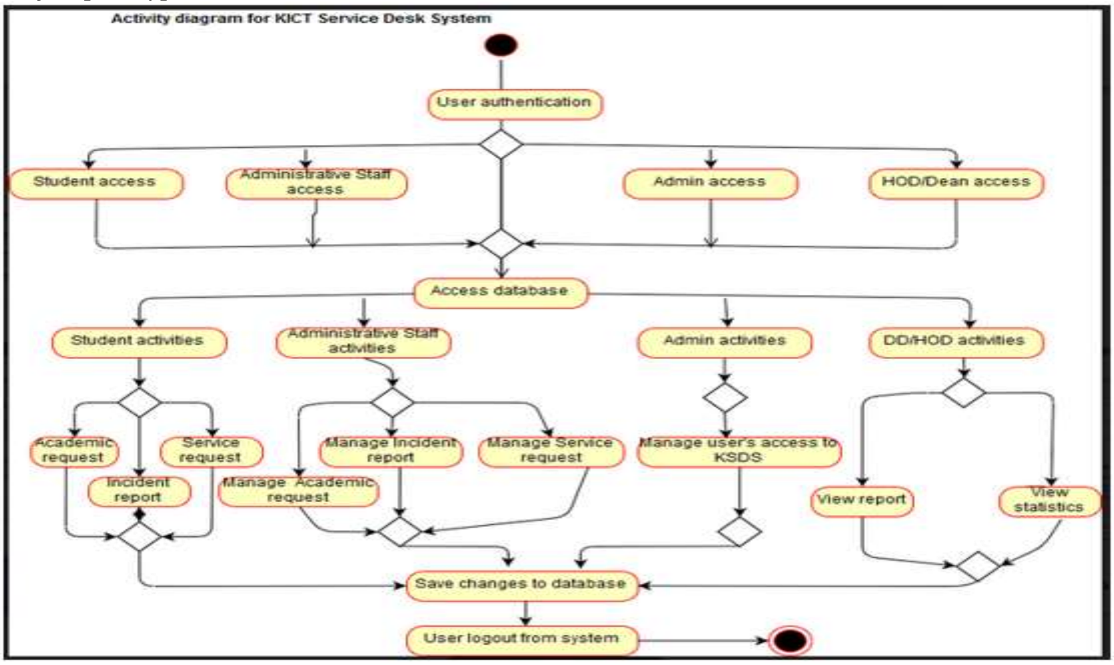

Fig 2. Activity Diagram for KICT Service Desk System

2) The KICT Service Desk System Use case Diagram

A use case diagram can be described as a representation of user's interaction with the system and representing the specifications of a use case. A use case diagram can be used in portraying the different types of users of a system and the different ways interaction is done with the system. The use case illustration for the proposed system is presented in Figure 3.

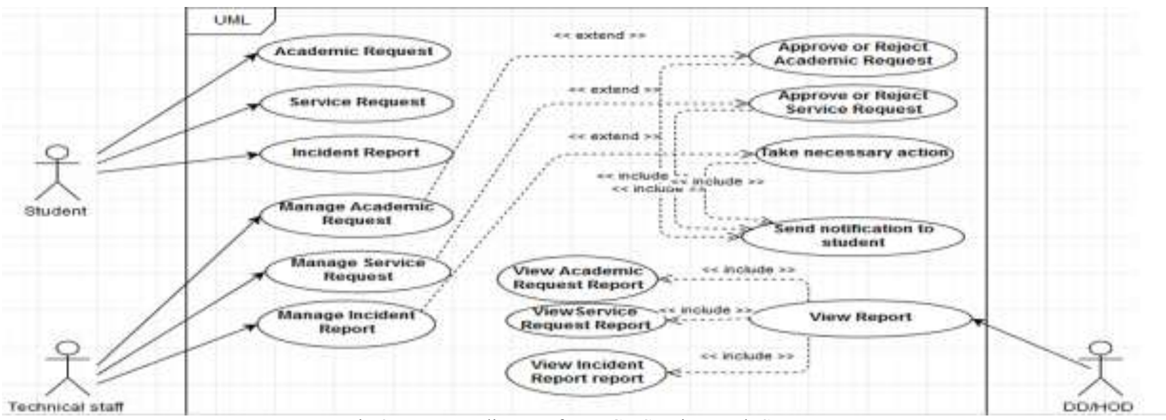

Fig 3. Use case diagram for KICT Service Desk System

As indicated in Figure 3, a Use Case describes the relationship between a system and its environment. A Use Case helped us to define a goal-oriented set of interactions between external actors and the system under consideration. There are overview and detail use cases. Overview use cases comprise the use case name, ID number, primary actor, type and a brief description.

3) Class Diagram

A class diagram can be described as a type of static structure diagram that shows and explains relationship between classes which are represented by boxes.

It has three(3) parts: top shows the name of the class; middle contains attributes and bottom gives the name of methods. Class diagram for this study is presented in Figure 4.

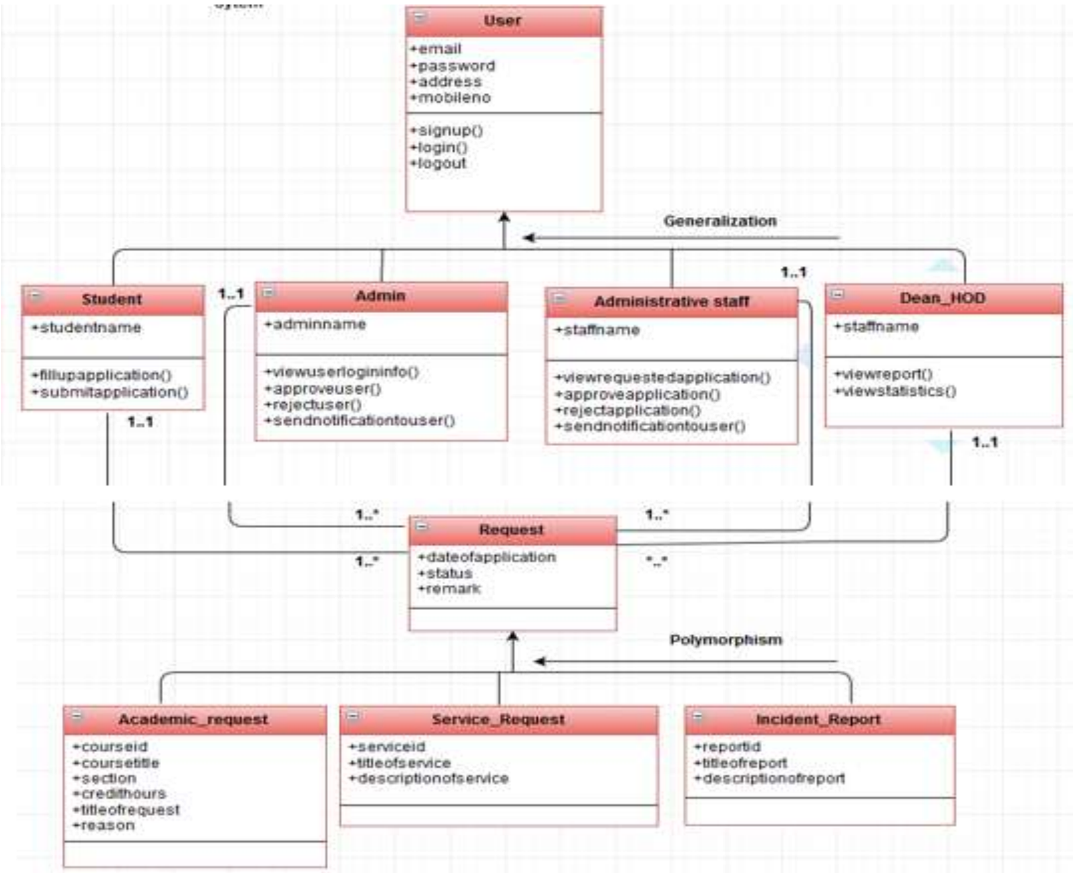

Fig 4. Class diagram for KICT Service Desk System

4) Database Design

Database design is an important part of the application development lifecycle. To be more precise, it is a methd of generating a detailed data model of a database. It is made of tables' columns, keys, indexes and relationships. In fact, it can be put into use by the overall process of designing, including the forms and queries used as part of the overall database application within the database management system (DBMS). Entity-Relationship model is a brief way of presenting a database. In relational database management system, data stored in one table can point to other tables. The data flow model for this study is presented in Figure 5.

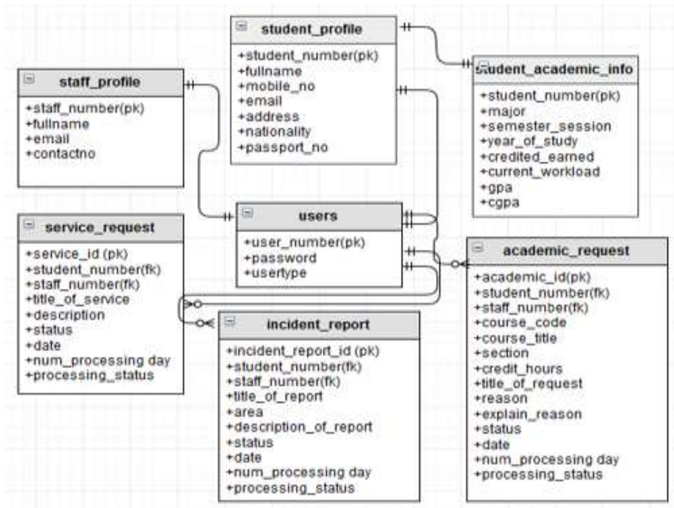

Fig 5. Database diagram (ERD) for KICT Service Desk System

C. Materials and Method

The programming language used in this study is Hypertext Markup Language (HTML), PHP with the support of Cascading Style Sheets (CSS3) and JavaScript. For the database for this system, MySQL will be utilized as the main framework to create, maintain and store all the data required for this system. The software specifications are: Adobe Photoshop CS5.5, PHP, Hypertext Markup Language, Bootstrap, Cascading Style Sheets (CSS), JavaScript, Database – MySQL Server, phpMyAdmin Browser - Internet Explorer 10 & above, Firefox 3.1.3 & above, Google Chrome 17 & above, Opera 17 & above, Hardware requirements – Workstation, Broadband, Laptop and Printer.

III. IMPLEMENTATION AND EVALUATION

The development of the proposed system came after a series of system development by the same research group under different titles such as [14 – 18]. In this system, there are three main components in the implementation consisting of the system integration, system output and system testing. KICT Service Desk System is a customized system that has been built to make the job more efficient and systematic. A paperless environment is the main reason why the system is developed. The system integration is the combination of all small parts of the functions. There is a common login page for all users. If the user is already registered, they would be able login to the system by using their matric number or staff number and password without any error. However, if the student is not registered to the system, they can easily do the registration by providing their details (full name, matric no, iium email address) in the registration page. Figure 6 is the registration page for student. However, student registration will be completed once admin approves the student to the system from the admin system. Once admin approves the student credentials, student will receive a notification through email with student's details and temporary password. By using the matric number and temporary password student can login to the system. After login to the system, the student can change his/her temporary password. There is no page for administrative staff and top management to do the registration through the system. They have to send their details (full name, staff no, email address) and type of login privilege to admin through email. Admin will check staff's credentials with the respective department and will complete the registration for staff with user's privilege. Once the admin has completed the registration for staff, the staff will receive a notification with staff's details and temporary password. Then the staff can login to the system by using staff number and temporary password. After login to the system, the staff can change his/her temporary password.

Fig 6. Student's main interfaces

The system output is based on user's category (Student, administrative staff, Top management and Admin). Once the student has successfully login to the system, it will display the main interface of the system. The main interfaces are: Student's name (profile), Dashboard, Academic Request, Service Request, Incident Report. Student can view his/her profile as shown in Figure 7 and student can change his/her temporary password from there.

Fig 7. Student's profile

The administrative dashboard provide access to view all academic request record like "My Academic Request Record", Admin can view My Service Request Record and My Incident Report Record made by him. When student browse View in My Academic Request Record student can view the details for that request. From there, student can download his/her requested application by clicking on download button. It will explain briefly how administrative staff can process student's request. Once administrative staff successfully log in to the system, it will display the main interface of the system. The main interfaces are: Staff's name (profile), Dashboard, Academic Request / Service Request, Incident Report There are two types of administrative staff. "staff academic" who will handle and process the academic request and "staff service" who will handle and process the service request. Both types of staff can handle incident report. When administrative staff browse the staff's name, he/she can view his/her profile and can also update his/her profile and change his/her temporary password from there. When academic staff browse Academic Request, he/she can view all the request title of Academic with notification of pending request. Likewise, for service staff service. However, staff can choose any request title. After choosing any request title, all the request will display for staff to be processed. Whenever staff browse View, he/she will view full application, and can accept or reject the application. After accepting or rejecting application, system will send a notification to student with student's details and application status.

The top management can view students request and monitor the performance of administrative staff. Once top management successfully login to the system, it will display the main interface of the system. The main interfaces are: Top management staff's name (profile), Dashboard, Academic Request, Service Request, Incident Report. Like in administrative staff. Top management can browse his/her profile, update his/her profile and change his/her temporary password. When top management browse dashboard, the system will display student's request summary and also how many request process on time and late. However, top management can view Academic Request summary in Academic Request section, Service Request summary in Service Request section and Incident Report summary in Incident Report section. Admin can approve and decline students and staff's login access to KICT service Desk System. Once admin successfully login to the system, it will display the main interface of the system. The main interfaces are: Admin's name (profile), User ID Activation, Staff Registration, User login Info. Like administrative staff. Admin can browse his/her profile, update his/her profile and change his/her temporary password. In User ID Activation interface, admin can approve and reject students request to access to KICT Service Desk System.

IV. CONCLUSION

This study developed has a prototyped capable of given an interactive theme for service support. The system is tested; it was written in a series of individual modules. Each module is functioning well. The testing ensures that interfaces between the modules work properly (integration testing), the system works on the intended-platform and that the system does what the user requires (Acceptance beta testing). From the testing, users were satisfied with the system. According to them, the functions and features are quite good, because the system is more flexible, reliable and improve administrative task of KICT. Future enhancement should dwell on new activities that might come up later. Thus, the whole process can go through a constructive conversation so that both the students and the admins would be notified if needed. Top management of the Kulliyah can view the summary of the service progression to inculcate the new upcoming tasks. The significant of the results of this study is the provision of interactive information-based systems where user involvement is put at the center. This study is designed as a conceptual framework of SDS systems that could increase the work performance to be more competent, accurate, consistent and reliable to suit with the current user's task. This will help the individual in handling their tasks. Although there are a lot of institutions that still prefer to use a manual system to do their work, but with this interactive SDS proposed in this study, it can aid the in the collection, tracking and processing requests. The system is dynamic where KICT SDS can cover various users' tasks.

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